Notes: Luo (JF, 2005)

Do Insiders Learn from Outsiders? Evidence from Mergers and Acquisitions

Contents

1 Big picture

- **Question.** Do managers learn from the stock market when deciding whether to complete an announced merger or acquisition?
- **Core idea.** In many M&A deals, managers first privately negotiate and announce the transaction. Outside investors then react through trading. The paper asks whether managers treat that price reaction as information and incorporate it into the subsequent closing decision.
- **Meaning of learning.** Learning means an information flow from outsiders to insiders: the market reaction to the announcement contains information not already known by managers, and managers use that information before completing or cancelling the deal.
- **Why M&A is a useful setting.** M&A announcements are discrete, visible corporate investment decisions. After the announcement but before closing, the firm has a second observable action: consummation, cancellation, or renegotiation.
- **Main empirical result.** The combined bidder-target announcement return predicts whether the deal is later completed. Deals with more favorable market reactions are more likely to close.
- **Key identification problem.** This raw relation is not enough to prove learning, because market prices may already reflect the market's expected probability of completion and because better deals may be both more positively received and more likely to complete.
- **Methodological solution.** The paper constructs a market-estimated synergy variable and tests whether the link between synergy and completion is stronger in subsamples where managers should have stronger incentives to learn.
- **Main cross-sectional evidence.** Managers appear more responsive to market signals in pre-agreement deals, in non-high-tech deals, and when bidders are smaller.
- **Renegotiation evidence.** The bidder's announcement return also predicts later offer revisions, consistent with market feedback influencing not only completion but also bargaining terms.
- **Economic takeaway.** The stock market is not only a passive evaluator of corporate decisions. In this setting, prices appear to transmit information back to firms and affect real corporate actions.

2 Incremental contribution of the paper

- **From market reaction to managerial action.** Earlier event studies document market reactions to M&A announcements. Luo asks whether those reactions influence later corporate decisions.
- **Evidence on information flowing from markets to firms.** The paper contributes to the literature on feedback from prices to real decisions. It provides large-sample evidence that insiders may learn from outsiders.
- **A new empirical design for learning.** The paper emphasizes that a simple correlation between announcement returns and completion is not enough. It develops a test that separates learning from probability-feedback and common-information explanations.

- **Use of market-estimated synergy.** The paper replaces raw announcement returns with a proxy for expected merger synergy, estimated by adjusting announcement returns for expected completion probabilities.
- **Heterogeneity as identification.** The paper tests whether the market signal matters more when learning should be more valuable: when cancellation is cheaper or when the market is relatively more informative.
- **Distinguishing deal-level traits.** Announcement timing, high-tech status, and bidder size are used to proxy for cancellation costs and relative information advantages.
- **Extension to renegotiation.** The paper shows that market feedback may also affect offer revisions, broadening the interpretation from deal completion to deal terms.
- **Corporate finance implication.** The paper links market efficiency to internal investment efficiency: prices can discipline or inform real decisions before irreversible commitment.

3 Research question and economic mechanism

3.1 Why managers might learn from the market

- M&A decisions are made by a small group of insiders: senior executives, boards, lawyers, bankers, and advisers.
- Outside investors may have information about industry prospects, macroeconomic conditions, competitive reactions, financing conditions, or valuation errors that insiders have missed.
- Analysts and institutional investors can process public information more broadly than the deal team.
- If the market reaction is strongly negative, managers may infer that the deal destroys shareholder value or that they overlooked relevant risks.

Interpretation: Learning is most plausible when the market is not simply reacting to insider information but aggregates outside information that can improve the firm's decision.

3.2 Why managers might not learn

- Managers may have private information that dominates the market's information.
- Managers may suffer from hubris or agency problems, making them ignore shareholder reactions.
- Cancelling a deal can be legally costly, reputationally costly, or strategically costly.
- Understanding the market reaction is itself costly because managers must communicate with investors and separate deal information from unrelated stock-price noise.

Interpretation: The learning hypothesis does not require every manager to learn. It predicts that learning is stronger when its benefits are higher or its costs are lower.

4 Data and measurement

4.1 M&A sample construction

- The initial sample comes from the SDC Domestic M&A Database.

- The sample period is January 1, 1990 to December 31, 1999.
- The deal types initially include disclosed-value M&As, leveraged buyouts, tender offers, and exchange offers.
- The target and acquirer must be U.S. public companies.
- The acquirer must seek to own at least 50% of the target after the transaction.
- The deal value must be at least $10 million.
- Hostile deals, unsolicited offers, unilateral proposals, competing bids, challenged bids, multilateral deals, self-tenders, asset sales, division sales, bankruptcy processes, and deals with multiple equity classes are excluded.
- The final full sample contains 2,114 friendly domestic M&A deals with usable CRSP announcement returns.

Interpretation: The sample is deliberately restricted to friendly one-bidder settings so that the completion decision can be interpreted as the joint decision of the merging parties, rather than as the outcome of takeover contests or hostile bargaining.

4.2 Announcement returns

- The bidder and target cumulative abnormal returns are measured over trading days $[-1, +7]$ around the announcement.
- Abnormal returns are based on a market model estimated over trading days $[-250, -50]$ before the announcement.
- The combined deal return, denoted DCAR, is the value-weighted average of the bidder and target announcement CARs.
- The paper uses DCAR as a market reaction to the deal as a whole.
- The bidder's announcement CAR is often negative, while the target's announcement CAR is usually positive.

Interpretation: The target's gain partly reflects takeover premia and bargaining. The combined deal return is closer to total surplus creation, which is why it is central to the paper.

4.3 Key deal traits

- **EarlyAnn.** Equals one for pre-agreement deals and zero for agreement deals. Pre-agreement deals are announced before a definitive merger contract is signed.
- **LowTech.** Equals one for non-high-tech deals and zero for high-tech deals.
- **BMV.** Bidder market value, measured before the announcement. Smaller bidders are expected to have less in-house valuation capacity and thus greater potential to learn from the market.

Interpretation: These traits are not merely controls. They are central identifying variation because they proxy for the incentive to learn.

4.4 Outcome variables

- **COMPLETE.** A dummy equal to one if the announced deal is completed and zero if it is cancelled.
- **Renegotiation.** For completed deals announced after January 1, 1995, a deal is classified as renegotiated if SDC reports amended contract terms and a change in offer price.
- **Offer revision direction.** Renegotiated deals are split into increased-offer and reduced-offer cases.

5 Empirical method

5.1 Why raw announcement returns are not enough

A naive test would estimate whether announcement returns predict deal completion. This is informative but not decisive. Luo emphasizes two non-learning explanations.

5.1.1 Probability-feedback

- The market reaction to a deal announcement may already reflect the market's expectation that the deal will or will not close.
- Suppose two deals have the same expected synergy if completed, but one is expected to close with probability 50% and the other with probability 100%.
- The deal expected to close for sure will have a larger announcement return even without learning.
- Therefore, announcement returns may predict completion simply because expected completion probability is embedded in the price reaction.

Interpretation: Raw returns confound expected value conditional on completion with expected probability of completion.

5.1.2 Common information

- Some deals are obviously better than others based on information already known to both managers and the market.
- Better deals should have higher announcement returns and are also more likely to close.
- This creates a positive return-completion relation even if managers ignore the market reaction.

Interpretation: Common information generates correlation without feedback from prices to managerial decisions.

5.2 Market-estimated synergy

The paper constructs a proxy for expected deal synergy, denoted **SYNERGY**. Conceptually, SYNERGY is the market's estimate of the total value created if the deal is completed with certainty. The relation is approximated as

$$\text{DCAR} \approx \Pr(\text{completion}) \times \text{SYNERGY}. \tag{1}$$

Thus,

$$\text{SYNERGY}^{(i)} = \frac{\text{DCAR}}{\text{Pr}^{(i-1)}}. \tag{2}$$

The completion probability is estimated iteratively using a probit model:

$$\text{COMPLETE} \sim \gamma_c^{(i)} + \Delta\gamma^{(i)}\text{TRAIT} + \eta^{(i)}\text{SYNERGY}^{(i)} + \iota^{(i)}Q, \tag{3}$$

with

$$\overset{(i)}{\text{Pr}} = \Phi\left(\gamma_c^{(i)} + \Delta\gamma^{(i)}\text{TRAIT} + \eta^{(i)}\text{SYNERGY}^{(i)} + \iota^{(i)}Q\right). \tag{4}$$

Interpretation: This adjustment tries to remove the probability-feedback component of raw announcement returns.

5.3 Core learning test

The main probit specification is

$$\text{COMPLETE} \sim \alpha_c + \Delta\alpha\,\text{TRAIT} + b_c\,\text{SYNERGY} + \Delta b\,(\text{TRAIT} \times \text{SYNERGY}) + cQ. \tag{5}$$

- The key coefficient is Δb.
- Under learning, SYNERGY should predict completion more strongly in samples where managers have a greater incentive to learn.
- Under no-learning, the common-information effect should not produce the predicted cross-sectional pattern in Δb.
- Q contains controls such as pre- and post-announcement stock returns, deal value, deal value relative to bidder size, same-state indicator, and cash-payment indicator.

Interpretation: The design uses heterogeneity in learning incentives to distinguish learning from simple market forecasting.

6 Theoretical model

6.1 Timeline and agents

- The shareholder S represents the market.
- The manager M represents the merging firms.
- At $t = 1$, the manager estimates deal synergy and decides whether to announce the deal.
- At $t = 2$, the market observes the announcement and estimates synergy.
- At $t = 3$, the manager observes the market reaction and may update his valuation.
- At $t = T$, the manager makes the completion decision.

Manager utility is written as

$$U_t = z + V_{Mt}, \tag{6}$$

where V_{Mt} is the manager's estimate of synergy and z captures agency benefits, private benefits, or hubris.

Interpretation: Positive z means managers may favor the merger even if the deal is not value-maximizing for shareholders.

6.2 Information structure

The model splits information into three components around announcement:

- x_0: common signal known to both managers and the market.
- x_1: manager-exclusive signal.
- x_2: market-exclusive signal.

Before learning, the manager's estimate is

$$V_{M1} = X + x_0 + x_1, \tag{7}$$

while the market's estimate is

$$V_{S2} = X + x_0 + x_2. \tag{8}$$

If the manager learns perfectly from the market, then

$$V_{M3}^{L} = X + x_0 + x_1 + x_2. \tag{9}$$

If the manager does not learn,

$$V_{M3}^{N} = X + x_0 + x_1. \tag{10}$$

Interpretation: The market's private signal is what can make the stock-price reaction useful to insiders.

6.3 Proposition 1: learning raises the slope of completion on synergy

The model implies that in a probit regression of completion on market-estimated synergy,

$$\text{COMPLETE} \sim a + b \times \text{SYNERGY}, \tag{11}$$

the slope is larger for learners than for non-learners:

$$b^{L} > b^{N}. \tag{12}$$

For learners, the market's signal enters the manager's closing valuation. For non-learners, the market-exclusive component of SYNERGY is an error-in-variable that attenuates the slope.

Interpretation: If managers use the market signal, the market's estimated synergy should have more explanatory power for the final completion decision.

6.4 Proposition 2: when learning is more valuable

The model gives the value of the market's unique signal to the manager as

$$K = \sigma_2 \phi\left(-\frac{W}{\sigma_2}\right) - W\Phi\left(-\frac{W}{\sigma_2}\right), \tag{13}$$

where $W = C + z + V_{M1}$, C is the cancellation cost, and σ_2 measures the informativeness of the market's unique signal. The comparative statics are

$$\frac{\partial K}{\partial C} < 0, \qquad \frac{\partial K}{\partial \sigma_2} > 0. \tag{14}$$

- Learning is more valuable when cancelling the deal is cheaper.
- Learning is more valuable when the market has more unique information.
- Learning becomes less valuable when agency or hubris benefits are very high.

Interpretation: These comparative statics justify the paper's three empirical hypotheses.

7 Hypotheses and identification

7.1 H1: announcement timing

- **Hypothesis.** Companies are more likely to learn in pre-agreement deals than in agreement deals.
- **Reason.** Cancelling an agreement deal violates a formal contract and may trigger legal, reputational, and strategic costs.
- **Prediction.** The coefficient on EarlyAnn $\times$ SYNERGY should be positive.

7.2 H2: high-tech opacity

- **Hypothesis.** Companies are more likely to learn in non-high-tech deals than in high-tech deals.
- **Reason.** High-tech deals often depend on proprietary technical information that firms do not disclose to investors; outside investors therefore have less useful information.
- **Prediction.** The coefficient on LowTech $\times$ SYNERGY should be positive.

7.3 H3: bidder size

- **Hypothesis.** Smaller bidders are more likely to learn from the market than larger bidders.
- **Reason.** Small firms have fewer internal analytical resources, less specialized managerial expertise, and often less access to sophisticated advisory capacity.
- **Prediction.** The coefficient on BMV $\times$ SYNERGY should be negative because higher BMV means a larger bidder and weaker learning incentive.

7.4 What is identified

- The empirical design identifies whether completion decisions are more sensitive to market-estimated synergy when theory predicts stronger learning incentives.
- It does not directly observe managerial meetings with investors or board discussions.
- The evidence is therefore behavioral: the market signal predicts later choices in ways not easily explained by non-learning alternatives.

8 Tables and figures

8.1 Figures 1 and 2: Timing and information structure

Read:

- Figure 1 shows four decision points: manager valuation before announcement, market valuation after announcement, manager valuation after seeing the market reaction, and final closing.
- Figure 2 decomposes information into manager-exclusive information, common information, and shareholder-exclusive information.
- The market's exclusive signal is the key source of learning.

Economic message: The learning mechanism requires two ingredients: a market signal not already known by managers and a second firm action after the signal is observed.

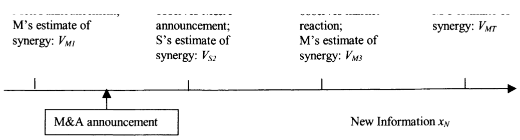

Figure 1: Timeline of events

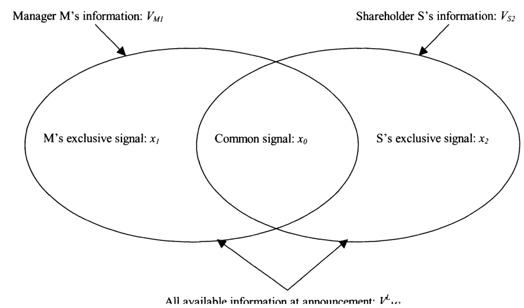

Figure 2: Information distribution after the announcement

8.2 Tables I and II: Sample construction and announcement returns

Read:

- Table I starts from 36,001 SDC domestic M&A observations and applies filters to obtain 2,114 friendly U.S. public-bidder/public-target deals with usable CRSP announcement returns.
- Major exclusions remove hostile deals, competing bids, challenged offers, unclear announcement timing, self-tenders, asset sales, bankruptcies, and multilateral deals.
- Table II reports that DCAR averages 0.8%, the bidder announcement CAR averages -1.6%, and the target announcement CAR averages 17.4%.
- Only 42.4% of bidder CARs are nonnegative, while 81.5% of target CARs are nonnegative.

Economic message: The combined return is more appropriate than either side's return alone because the paper is interested in total deal surplus, not bargaining transfers between bidder and target.

Table I

Sample Selection

This table reports my sample-selection process. My initial sample of mergers and acquisitions (M&As) is from the SDC Domestic M&A database (SDC). I also supplement data from SDC with data from the Dow Jones Interactive (DJI). In the table, the first column contains the number of observations after each query step, and the second column describes the query step. To obtain the cumulative abnormal return (CAR) of a stock, I estimate the following market model:

$$R_{it} = a_i + \beta_i R_{Mt} + \varepsilon_{it},$$

where R_{it} is the daily stock return, R_{mt} is the daily return of the value-weighted CRSP market portfolio, and ε_{it} is the daily abnormal return. All stock return data are from CRSP. I estimate the model between trading days −250 to −50 before the M&A announcement. I set the bidder's α at zero because the bidder often outperforms prior to the announcement. The target's α is unconstrained.

No. of Obs. After Query	Query Description
	Machine search in SDC
	SDC Domestic M&As announced between 1/1/1990 and 12/31/1999
36,001	Deal type included: disclosed value M&As, leveraged buyouts, tender offers, and exchange offers
17,557	Percent of shares acquirer is seeking to own after transaction: 50% or higher
17,556	Target nation: United States
5,479	Target is public
4,982	Acquirer nation: United States
3,602	Acquirer is public
3,235	Deal value is $10 million or higher
	Hand collecting data in DJI and SDC
2,569	Exclude:
	(1) Hostile target initial reaction, unilateral proposal, unsolicited offer, deal started with tender or block purchase from a third party
	(2) Competing bid
	(3) Offer is challenged while open
	(4) No news record or unclear record for announcement timing
	Two SDC announcement dates (SDC deal numbers: 514511020 and 390133020) are changed according to DJI
	(5) Announcement is deal rejection or completion or deal is still pending
	(6) Self tender, asset sale, division sale, bankruptcy process, exchange offer, bidder already a majority owner or exercise of an early option
	(7) Multilateral deal, or deal in which parties have multiple equity classes
2,133	Both bidder and target are identified in CRSP, matched by CUSIP, ticker and/or company name, and have [−1, +1] trading-day deal-announcement CARs
2,114	Deals have [−1, +7] trading-day deal-announcement CARs; the full sample

Table I: Sample selection

Table II

Descriptive Statistics of M&A Announcement Returns

This table describes the market reaction to the M&A announcement. I use the cumulative abnormal returns (CARs) of the merging companies around the announcement to represent the market reaction. I define the event window as the trading days [−1, +7] around the announcement. The deal return (DCAR) is the value-weighted average of the bidder's and the target's announcement CARs.

					Correlations with	
Variable	Mean	*SD*	*t*-Statistics	Frequency ≥ 0	Bidder Ann. CAR	Target Ann. CAR
DCAR	0.8%	8.7%	4.3	54.4%	0.86	0.55
Bidder's announcement CAR	−1.6%	9.0%	−8.0	42.4%		0.17
Target's announcement CAR	17.4%	23.3%	34.4	81.5%		

Table II: Descriptive statistics of announcement returns

8.3 Tables III and IV: Preliminary completion evidence

Read:

- Table III shows that completed deals are more common when DCAR is nonnegative.
- In the full sample, completion is 93.0% when DCAR is nonnegative and 90.6% when DCAR is negative.
- The difference is much larger in pre-agreement deals: 80.7% versus 64.1%, a 16.6 percentage point gap.
- The completion gap is also larger for non-high-tech deals than high-tech deals and larger for small bidders than large bidders.
- Table IV confirms the preliminary relation in probit regressions with controls.
- DCAR predicts completion in the full sample and has stronger coefficients in the subsamples where learning should be stronger.

Economic message: The raw evidence already looks like learning, but it is not conclusive because DCAR can reflect anticipated completion probability and common deal quality.

Table III

M&A Completion Frequency and the Sign of Market Reaction to the M&A Announcement

This table reports the relation between the sign of the combined deal return (DCAR) and the deal completion frequency. The table shows the relation in the full sample and in seven subsamples: pre-agreement deals (EarlyAnn = 1), agreement deals (EarlyAnn = 0), non-high-tech deals (LowTech = 1), high-tech deals (LowTech = 0), deals with small bidders ($2.3 million < BMV < $515.4 million), deals with medium bidders ($515.8 million < BMV < $2,664 million), and deals with large bidders ($2,698 million < BMV < $505,796 million). If two companies announce their M&A after they have signed a definitive merger contract, the deal is an agreement deal. Otherwise, it is a pre-agreement deal. If both the bidder and the target are high-tech companies, the deal is a high-tech deal. Otherwise, it is a non-high-tech deal. I use the bidder's and the target's high-tech status codes in SDC to judge whether they are high-tech companies. The variable BMV is the bidder's market capitalization at the close of the trading day −41 before the announcement. The three bidder size categories each have the same number of observations. I use the sign rather than the magnitude of DCAR because, intuitively, companies should complete the deals that increase shareholder value, no matter how much the increase is. In the table, the completion frequency is in percentage, and the number of observations is in parentheses. The last row of the table reports the difference in completion frequency between the deals with DCAR ≥ 0 and those with DCAR < 0. The *p*-values of the differences are in italics.

	Full Sample	Announcement Timing		High-Tech Deal Classification		Bidder Stock Market Capitalization		
	(a)	(b) Pre-agreement	(c) Agreement	(d) Non-high-tech	(e) High-Tech	(f) Small	(g) Medium	(h) Large
Unconditional	91.9%	73.7%	95.3%	92.1%	91.2%	87.2%	93.0%	96.2%
(Obs.)	(2114)	(334)	(1780)	(1660)	(454)	(674)	(675)	(675)
DCAR ≥ 0	93.0%	80.7%	95.5%	93.4%	91.7%	89.6%	94.5%	96.0%
(Obs.)	(1149)	(192)	(957)	(933)	(216)	(385)	(364)	(351)
DCAR < 0	90.6%	64.1%	95.1%	90.5%	90.8%	84.1%	91.3%	96.3%
(Obs.)	(965)	(142)	(823)	(727)	(238)	(289)	(311)	(324)
Difference = (DCAR ≥ 0) − (DCAR < 0)	**2.5%****	**16.6%*****	**0.4%**	**2.9%****	**0.9%**	**5.5%****	**3.2%**	**−0.3%**
p-value	*0.038*	*0.001*	*0.714*	*0.033*	*0.733*	*0.033*	*0.105*	*0.848*

***, **, *: Statistically significant at the 1%, 5%, and 10% levels, respectively.

Table III: Completion frequency and sign of market reaction

Table IV

Probit Regressions of the M&A Closing Decision on the Market Reaction to the M&A Announcement

This table reports nine probit regressions of the M&A closing decision on the market reaction to the M&A announcement controlling for deal characteristics. The dummy COMPLETE equals 1 for completed deals and 0 for cancelled deals. Each regression corresponds to a specific sample and model combination. I measure the bidder's and the target's post-announcement CARs from the trading day +8 after the announcement to the trading day −2 before the closing. I measure the two companies' pre-announcement CARs over the trading days [−40, −2] before the announcement. I obtain from SDC the data for Deal value, whether the bidder and the target are headquartered in the same state, and whether the exchange medium is pure cash. For each test, the number of observations used in the regression is in parentheses, and the *p*-values of the coefficients are in italics.

	Full Sample—Two Tests		Announcement Timing		High-Tech Deal Classification		Bidder Equity-Market Capitalization		
	(a)	(b)	(c) Pre-Agreement	(d) Agreement	(e) Non-High-Tech	(f) High-Tech	(g) Small	(h) Medium	(i) Large
Constant	1.53***	1.52***	0.770***	1.80***	1.53***	1.57***	1.08***	1.76***	2.11***
	0.000	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*
Deal announcement CAR (DCAR)	**1.35*****		**4.20*****	**0.618**	**2.04*****	**0.204**	**1.51****	**2.37****	**−0.544**
	0.005		***0.000***	***0.304***	***0.001***	***0.806***	***0.014***	***0.012***	***0.696***
Bidder's announcement CAR		**1.02****							
		0.027							
Target's announcement CAR		**0.168**							
		0.390							
Bidder's post-announcement CAR	0.611***	0.616***	1.07**	0.663**	0.765***	−0.121	1.12***	0.291	0.283
	0.006	*0.006*	*0.033*	*0.016*	*0.002*	*0.816*	*0.004*	*0.522*	*0.512*
Target's post-announcement CAR	1.22***	1.22***	1.00**	1.21***	1.39***	1.18**	0.500	2.59***	1.24*
	0.000	*0.000*	*0.027*	*0.003*	*0.000*	*0.027*	*0.212*	*0.000*	*0.090*
Bidder's pre-announcement CAR	0.545*	0.514*	0.773	0.733*	0.521	0.652	0.802**	0.778	0.284
	0.071	*0.087*	*0.131*	*0.092*	*0.169*	*0.203*	*0.046*	*0.213*	*0.751*
Target's pre-announcement CAR	−0.150	−0.151	−0.357	−0.119	−0.0990	−0.327	−0.0828	−0.233	−0.688
	0.406	*0.409*	*0.252*	*0.632*	*0.653*	*0.318*	*0.736*	*0.531*	*0.131*
Deal value (billion $)	0.00276	0.00394	0.136	−0.0111	0.0286	−0.0115	1.01**	0.252	−0.00151
	0.816	*0.747*	*0.464*	*0.263*	*0.322*	*0.447*	*0.037*	*0.153*	*0.908*
Deal value/BMV	−0.123***	−0.114***	−0.133**	−0.111***	−0.128***	−0.162	−0.115***	−0.618**	−0.831***
	0.000	*0.000*	*0.011*	*0.001*	*0.000*	*0.113*	*0.003*	*0.011*	*0.000*
Bidder and target headquartered in same state (1 = yes)	0.250**	0.254**	0.0278	0.510***	0.264**	0.146	0.277**	0.514**	0.702*
	0.014	*0.012*	*0.878*	*0.008*	*0.022*	*0.513*	*0.044*	*0.017*	*0.086*
Pure cash deal (1 = yes)	−0.228**	−0.228**	−0.0460	−0.332**	−0.114	−0.552***	0.259	−0.231	−0.227*
	0.029	*0.029*	*0.824*	*0.011*	*0.356*	*0.007*	*0.109*	*0.244*	*0.332*
(Obs.)	(2004)	(2004)	(302)	(1702)	(1574)	(430)	(666)	(669)	(669)

***, **, *: Statistically significant at the 1%, 5%, and 10% level, respectively.

Table IV: Probit regressions of closing decision on market reaction

8.4 Table V: Core learning tests with probability-feedback and common-information controls

Read:

- Table V is the central empirical table.
- The dependent variable is COMPLETE.
- SYNERGY is estimated iteratively from DCAR and predicted completion probability.
- The key coefficients are the interaction terms: EarlyAnn × SYNERGY, LowTech × SYNERGY, and BMV × SYNERGY.
- Tests (a)-(c) examine each hypothesis separately.
- Test (d) includes all three interactions jointly.
- Test (e) substitutes DCAR for SYNERGY as a robustness check.
- The interaction coefficients have the predicted signs and are statistically significant or marginally significant.

Economic message: After controlling for probability-feedback and common-information concerns, market-estimated synergy matters more exactly where the model predicts stronger incentives to learn.

Table V

Tests of Learning Controlling for the Probability-Feedback and the Common-Information Problems

This table reports five probit regressions. The dependent variable is always COMPLETE, which equals 1 for completed deals and 0 for cancelled deals. The variable SYNERGY is the proxy for the market's estimate of synergy at the announcement assuming the deal will definitely be completed. In the tests (a)–(d), I estimate SYNERGY from the deal announcement return (DCAR) with the following iterative process:

$$\mathrm{SYNERGY}^{(i)} = \mathrm{DCAR}/\mathrm{Pr}^{(i-1)} \quad (\text{set } \mathrm{Pr}^{(0)} = 1), \tag{2}$$

$$\mathrm{COMPLETE} \sim \gamma_c^{(i)} + \Delta\gamma^{(i)}\,\mathrm{TRAIT} + \eta^{(i)}\,\mathrm{SYNERGY}^{(i)} + \imath^{(i)}\,\mathrm{Q}, \tag{3}$$

$$\mathrm{Pr}^{(i)} = \Phi\left(\gamma_c^{(i)} + \Delta\gamma^{(i)}\,\mathrm{TRAIT} + \eta^{(i)}\,\mathrm{SYNERGY}^{(i)} + \imath^{(i)} Q\right). \tag{4}$$

In these equations, $\Phi(\cdot)$ is the *c.d.f.* of the standard normal distribution. The sign "$\sim$" represents a probit regression relation. The variable TRAIT = {EarlyAnn, LowTech, BMV}; Q = {Bidder's pre-announcement CAR, Deal value, Deal value/BMV, Bidder, and target headquartered in same state (1 = yes), Pure cash offer (1 = yes)}; $\Delta\gamma$ and $\imath$ are also vectors. The superscript i is for the iteration step. Equation (2) calculates $\mathrm{SYNERGY}^{(i)}$ from DCAR and $\mathrm{Pr}^{(i-1)}$ (initial $\mathrm{Pr}^{(0)} = 1$). Equation (3) gives the probit regression coefficients. Equation (4) calculates $\mathrm{Pr}^{(i)}$ using the coefficients from equation (3). Only equation (3) needs estimation. Equations (2) and (4) simply calculate the left-hand variables. During the iteration, I censor $\mathrm{Pr}^{(i)}$ at the lower end at 0.00001 to avoid the division of zero. I also censor $\mathrm{SYNERGY}^{(i)}$ at −100% and +100%. The process continues until convergence.
For tests (a)–(d), at convergence ($i = 20$):

$$\mathrm{SYNERGY} = \mathrm{DCAR}/\mathrm{Pr}, \quad \text{and}$$

$$\begin{aligned}\mathrm{Pr} = \Phi\{&1.49230 - 1.07112\ \mathrm{EarlyAnn} + 0.23564\ \mathrm{LowTech} + 0.01485\ \mathrm{BMV(billion\ \$)}\\ &+ 1.46144\ \mathrm{SYNERGY} - 0.02849\,(\text{Deal value billion \$}) - 0.10608\,(\text{Deal value/BMV})\\ &+ 0.74705\,(\text{Bidder's pre-announcement CAR})\\ &+ 0.29232\,(1 \text{ for bidder \& target headquartered in same state, 0 otherwise})\\ &- 0.22244\,(1 \text{ for pure cash deals, 0 otherwise})\}.\end{aligned}$$

For test (e), I substitute DCAR for SYNERGY:

SYNERGY = DCAR.

The number of observations for the tests are in parentheses, and the *p*-values of coefficients are in italics.

	Test (a)^a	Test (b)^a	Test (c)^a	Test (d)^a	Test (e)^b
Constant	1.55***	1.54***	1.54***	1.54***	1.54***
	0.000	*0.000*	*0.000*	*0.000*	*0.000*
EarlyAnn (1 = pre-agreement deals)	−1.07***	−1.08***	−1.07**	−1.07**	−1.09**
	0.000	*0.000*	*0.000*	*0.000*	*0.000*
LowTech (1 = non-high-tech deals)	0.270**	0.291**	0.272**	0.276**	0.274**
	0.018	*0.010*	*0.017*	*0.015*	*0.016*
BMV (billion $)	0.0149**	0.0148**	0.0184***	0.0182***	0.0191***
	0.024	*0.025*	*0.008*	*0.009*	*0.008*
SYNERGY	0.623	0.0821	1.71***	−1.08	−0.162
	0.241	*0.908*	*0.000*	*0.890*	*0.854*
EarlyAnn × SYNERGY	**2.36****			**1.88****	**3.16****
	0.010			***0.046***	***0.014***

(*continued*)

Table V, part 1: Learning tests and iterative synergy construction

Table V—*Continued*

	Test (a)^a	Test (b)^a	Test (c)^a	Test (d)^a	Test (e)^b
LowTech × SYNERGY		**2.08****		**1.72***	**1.74****
		0.018		***0.058***	***0.010***
BMV × SYNERGY			**−0.170****	**−0.140***	**−0.153***
			0.034	***0.095***	***0.089***
Bidder's post-announcement CAR	0.708***	0.734***	0.717***	0.716***	0.705***
	0.003	*0.002*	*0.003*	*0.003*	*0.003*
Target's post-announcement CAR	1.17***	1.11***	1.11***	1.14***	1.16***
	0.000	*0.000*	*0.000*	*0.000*	*0.000*
Bidder's pre-announcement CAR	0.758**	0.743**	0.778**	0.777**	0.777**
	0.018	*0.020*	*0.014*	*0.016*	*0.015*
Target's pre-announcement CAR	−0.198	−0.178	−0.173	−0.192	−0.208
	0.307	*0.356*	*0.367*	*0.322*	*0.282*
Deal value ($billion)	−0.0312**	−0.0312**	−0.0288**	−0.0299**	−0.0313**
	0.017	*0.018*	*0.022*	*0.021*	*0.015*
Deal value/BMV	−0.108***	−0.121***	−0.119***	−0.117***	−0.109***
	0.002	*0.000*	*0.000*	*0.000*	*0.000*
Bidder and target headquartered in same state (1 = yes)	0.325***	0.315***	0.328***	0.328***	0.335***
	0.004	*0.005*	*0.003*	*0.004*	*0.003*
Pure cash offer (1 = yes)	−0.242**	−0.222**	−0.232**	−0.237**	−0.234**
	0.031	*0.049*	*0.038*	*0.035*	*0.037*
(Obs.)	(2004)	(2004)	(2004)	(2004)	(2004)

***, **, *: Statistically significant at the 1%, 5%, and 10% level, respectively.
[a]SYNERGY = DCAR/Pr, estimated iteratively according to equations (2)–(4).
[b]SYNERGY = DCAR.

and in deals with smaller bidders than those with larger ones. BMV in test (c) is a continuous variable. In essence, I test learning in deals with smaller bidders and use those with larger bidders as the control. The learning hypothesis H3 predicts that BMV × SYNERGY has a negative sign.[11,12]

[11] The completion frequency of pre-agreement deals is by far the lowest (74%) among the

Table V, part 2: Learning tests continued

8.5 Tables VI and VII: Renegotiation evidence

Read:

- Table VI examines which deal characteristics predict renegotiation among completed deals from 1995 to 1999.
- Pre-agreement deals are much more likely to be renegotiated, consistent with the idea that deals without definitive contracts are easier to revise.
- Table VII examines whether the market reaction predicts the direction of offer revision.
- In the ordered probit, the bidder announcement CAR is positive and marginally significant.
- Among renegotiated deals only, the bidder announcement CAR is strongly positive: when the bidder reacts more favorably, the offer is more likely to be revised upward; when the bidder reacts poorly, the offer is more likely to be revised downward.

Economic message: The renegotiation results extend the learning interpretation beyond closing versus cancellation. Market feedback also appears to influence deal terms.

Table VI

Renegotiation and M&A Deal Characteristics

This table reports the relations between the renegotiation decision and M&A deal characteristics. The sample consists of all completed deals announced between January 1, 1995 and December 31, 1999. The sample period has a later start point to avoid possible sample-selection problems in the renegotiation data from SDC. An M&A is "renegotiated" if (i) it is eventually completed, (ii) SDC reports that the contract terms are amended, and (iii) its final offer price is different from its initial offer price. The first column reports the correlations between the renegotiation dummy (one for renegotiated deals and zero for others) and deal characteristic variables. The second column reports a probit regression of the renegotiation decision on the deal characteristics. The number of observations for the tests are in parentheses, and the *p*-values of coefficients are in italics.

	(a) Correlations Variable = 1: Renegotiated Deals, 0: Others	(b) Probit Regression Dependent Variable = 1: Renegotiated Deals, 0: Others
Constant		−1.64***
		0.000
EarlyAnn (1 = pre-agreement deals)	0.12***	0.668***
	0.000	*0.000*
LowTech (1 = non-high-tech deals)	0.04	0.095
	0.156	*0.483*
BMV (billion $)	−0.02	−0.00024
	0.520	*0.923*
Deal value (billion $)	−0.03	−0.034
	0.258	*0.309*
Deal value/BMV	−0.02	−0.065
	0.413	*0.489*
Bidder and target headquartered	−0.01	−0.059
in same state (1 = yes)	*0.667*	*0.621*
Pure cash deal (1 = yes)	0.02	0.140
	0.361	*0.307*
Price collar (1 = yes)	0.03	0.172
	0.292	*0.228*
(Obs.)	(1458)	(1390)

***, **, *: Statistically significant at the 1%, 5%, and 10% level, respectively.

Table VI: Renegotiation and deal characteristics

Table VII

Renegotiation and Market Reaction

This table reports two probit regressions of the renegotiation decision on the market reaction to the M&A announcement. Test (a) is an ordered probit test with a three-level dependent variable, which is 1 for renegotiated deals with increased offers, 0 for not-renegotiated deals, and −1 for renegotiated deals with decreased offers. Test (a) has two intercepts because of the three levels of the dependent variable. The sample of test (a) consists of all completed deals announced between January 1, 1995 and December 31, 1999. Test (b) is an ordinary probit test with a two-level dependent variable, which is 1 for renegotiated deals with increased offers and 0 for renegotiated deals with decreased offers. The sample of test (b) only consists of completed and renegotiated deals announced between January 1, 1995 and December 31, 1999. Both tests include 94 renegotiated deals, among which the offer price increases in 42 deals and decreases in the other 52 deals. In both tests, I use the bidder's and the target's announcement CARs to represent the market reaction to the M&A announcement. The number of observations for the tests are in parentheses, and the *p*-values of coefficients are in italics.

Dependent Variable	Test (a)[a]	Test (b)[b]
Intercept	−1.94***	−0.0762
	0.000	*0.690*
Intercept2	3.81***	
	0.000	
Bidder's Announcement CAR	**0.895***	**9.01*****
	0.089	***0.001***
Target's Announcement CAR	**0.0414**	**1.45**
	0.846	***0.104***
Bidder's post-announcement CAR	0.588**	3.23**
	0.014	*0.016*
Target's post-announcement CAR	0.793***	5.03***
	0.004	*0.001*
Bidder's pre-announcement CAR	0.0214	0.0363
	0.945	*0.975*
Target's pre-announcement CAR	−0.0587	0.0846
	0.775	*0.916*
Sample	All completed deals	Completed and renegotiated deals
(No. of obs.)	(1457)	(94)

***, **, *: Statistically significant at the 1%, 5%, and 10% level, respectively.
[a] 1: Offer higher, 0: No change, −1: Offer lower.

Table VII: Renegotiation and market reaction

9 Main empirical findings

9.1 Raw market reaction predicts completion

- Deals with nonnegative DCAR are more likely to complete than deals with negative DCAR.
- This relation is economically meaningful because the unconditional cancellation rate is low, around 8.1% in the full sample.
- The market signal explains a nontrivial fraction of cancellation risk, especially in pre-agreement deals.

9.2 Learning is stronger when cancellation is easier

- Pre-agreement deals show a much stronger relation between market reaction and completion.
- Agreement deals have very high completion rates regardless of market reaction.
- This supports the idea that learning has higher value when managers can still change course at lower cost.

9.3 Learning is weaker in high-tech deals

- The market's signal has less predictive power in high-tech deals.
- The interpretation is not that high-tech deals are unimportant, but that outside investors may lack access to proprietary technical information.
- When the market is less informed, managers have less reason to respond to its reaction.

9.4 Smaller bidders learn more

- The market reaction matters more for deals involving smaller bidders.
- Small bidders likely have fewer internal valuation resources.
- As bidder size rises, the interaction BMV $\times$ SYNERGY is negative, consistent with lower dependence on market feedback.

9.5 Renegotiation supports the learning channel

- Learning should affect not only whether a deal closes but also how terms are revised.
- The bidder's announcement return predicts later offer revision direction.
- Poor bidder reactions are associated with downward revisions; favorable bidder reactions are associated with upward revisions.

10 Identification and empirical interpretation

10.1 Strengths of the design

- The design recognizes and directly addresses two major omitted-mechanism concerns.
- The estimated SYNERGY variable reduces probability-feedback contamination.
- The interaction strategy uses theoretically motivated heterogeneity to distinguish learning from common information.
- The findings are consistent across univariate tests, probit models, the corrected SYNERGY tests, and renegotiation tests.

10.2 Remaining limitations

- The paper does not observe managerial deliberations directly.
- SYNERGY is estimated rather than directly observed.
- The classification of high-tech deals is a proxy for market information, not a direct measure of information asymmetry.
- Renegotiation analysis cannot fully control probability-feedback and common-information stories because renegotiation is inherently about bargaining, not only total synergy.
- The sample focuses on friendly U.S. public-firm M&A deals in the 1990s, so the conclusions may not generalize to hostile takeovers, private-firm deals, or periods with different merger-law environments.

10.3 Appropriate reading of the evidence

- The paper does not claim that all market reactions are correct.
- It does not claim that managers mechanically follow price movements.
- It claims that managers appear to extract information from the market reaction, especially in settings where learning is relatively valuable.

- The strongest evidence comes from the cross-sectional pattern of sensitivities, not from the simple raw correlation alone.

11 Short summary

- Luo studies whether managers learn from the stock market during M&A decisions.
- The sample contains 2,114 friendly domestic M&A deals announced from 1990 to 1999.
- The combined bidder-target announcement return predicts whether the deal is later completed.
- Raw return-completion correlations are not sufficient because of probability-feedback and common-information explanations.
- The paper constructs a market-estimated SYNERGY measure that adjusts announcement returns for expected completion probability.
- The core test asks whether the completion-synergy relation is stronger when managers should have stronger learning incentives.
- Evidence supports three predictions: learning is stronger in pre-agreement deals, stronger in non-high-tech deals, and stronger for smaller bidders.
- Renegotiation tests show that bidder announcement returns predict whether offers are revised up or down.
- The paper concludes that firms appear to use market prices as information in real corporate decisions.

12 Conclusion

12.1 Core conclusion

Luo provides evidence that insiders learn from outsiders in mergers and acquisitions. Managers appear to use the market reaction to an M&A announcement when deciding whether to complete the deal and, in some cases, how to revise the terms. The strongest evidence is not merely that announcement returns predict completion, but that this relation is stronger when learning should be more valuable.

12.2 Economic intuition

The stock market aggregates dispersed information. In M&A deals, that information can concern deal synergies, industry prospects, valuation mistakes, financing risks, and competitive effects. Managers may not know all of this information when the deal is announced. If cancelling or revising the deal is feasible, they have an incentive to interpret the price reaction and adjust their later decisions.

12.3 Incremental contribution revisited

The paper advances corporate finance in three ways. First, it provides evidence that stock prices can affect real investment decisions, not just reflect them. Second, it develops a testing framework that corrects for probability-feedback and common-information mechanisms. Third, it demonstrates that learning depends on institutional and informational frictions: contract status, opacity, and bidder resources matter.

12.4 Implications for corporate governance

The findings imply that market reactions can discipline or inform managerial decisions. Boards and managers should not mechanically follow short-run stock returns, but they should treat large negative reactions as potentially informative signals, especially when outside investors may have valuable information and when the deal is still reversible.

12.5 Implications for event studies

Event-study returns should not always be interpreted as terminal market judgments. When firms respond to market feedback, the announcement return becomes part of the information environment that shapes subsequent actions. This feedback loop complicates simple interpretations of announcement effects.

12.6 Limitations

The paper cannot directly observe the communication between managers and investors. It relies on a model-based estimate of SYNERGY and on deal characteristics as proxies for learning incentives. Its renegotiation evidence is suggestive but less clean than the completion evidence because offer revision depends on bargaining. The sample is also specific to friendly public M&A deals in the 1990s.

12.7 Takeaway

The enduring takeaway is that the market can act as an information intermediary for firms. In M&A, managers appear to learn more when the market is informative and when abandoning or revising the deal is easier. This makes the stock market not only a valuation mechanism but also a source of feedback for corporate decision-making.